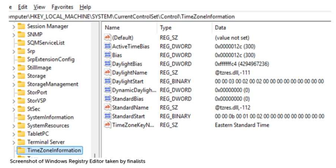
RegiMon: Monitoring the Windows Registry in Real-Time

PURPOSE

- Registry Run Keys (MITRE ATT&CK T1547.001) is linked to 54 distinct threat actor groups, making it the single most **widely documented persistence technique** in the MITRE ATT&CK framework (MITRE, 2024).
- Create a **scalable, lightweight, real-time** detection mechanism for Windows Registry modifications in memory
- Provide immediate user interaction, including the ability to **undo** suspicious changes at the moment of detection.

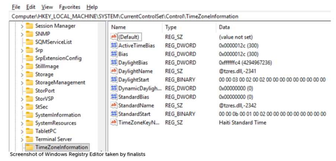
BACKGROUND

- The **Windows Registry** is a vast database that stores critical settings about the Windows operating system on a machine.
- These settings are in the form of **values** and are assigned to **keys** that are found in hives.
- There exists **two copies** of the Windows Registry, one that is located on-disk and one located **in-memory**.
- **Malicious** software can exploit the Windows Registry by **changing key values** or adding extra keys that can cause theft, damage, or deletion of data.
- Existing Solutions
  - RegRipper - Reads into a **singular** hive
  - RegShot - Takes snapshot of the **entire** registry
  - Volatility - Analyzes **ram-dumps**

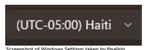


Screenshot of Windows Registry Editor taken by finalists

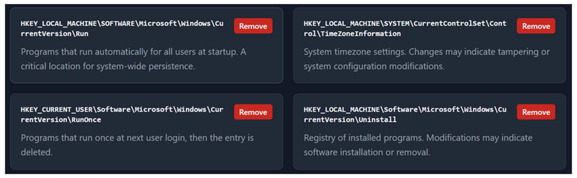
Screenshot of Windows Settings taken by finalists



Screenshot of Windows Registry Editor taken by finalists



Screenshot of Windows Settings taken by finalists

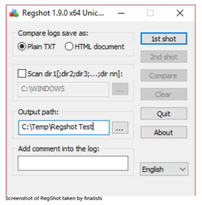


Screenshot of RegView taken by finalists

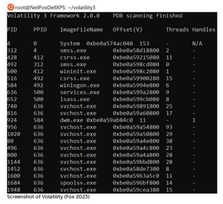
METHODOLOGY

Research

- Surveyed existing registry monitoring tools: RegRipper, RegShot, Volatility.
- Determined **RegNotifyChangeKeyValue()** as the optimal method for real-time detection.
- Researched mechanisms of malware hiding in RAM (Dolan-Gavitt).
- Investigated differences between hives in Microsoft documentation.



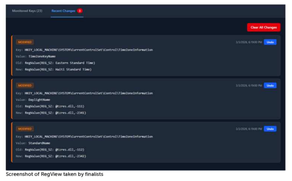
Screenshot of RegShot taken by finalists



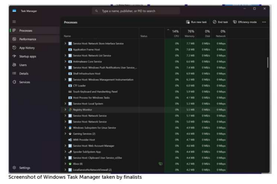
Screenshot of Volatility (Fox 2023)

Testing

- Attempted testing on ThinkCentre with ReactOS + Windows 7, finally settled on **Windows 10 VM**
- Manual registry edits via **regedit** (key addition, modification, deletion).
- System-level changes such as **timezone adjustments** to trigger HKLM writes.
- Executed **Love Bug virus** in isolated Windows 10 VM to validate.



Screenshot of RegView taken by finalists



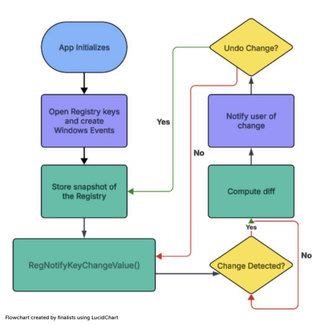
Screenshot of Windows Task Manager taken by finalists

SOFTWARE DESIGN

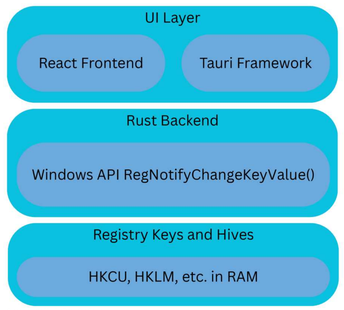
- Built in **Rust** for Windows libraries and **Tauri** framework
- **System tray** app in order to be unobtrusive and lightweight
- Uses **RegNotifyChangeKeyValue()** to listen for changes at the OS level — no polling, zero CPU overhead while idle
- Monitors HKCU without admin rights; HKLM accessible with elevated permissions
- Maintains a snapshot of previous registry state to compute **exact** diffs on change
- Stores monitored key paths persistently so preferences survive app restarts
- Upon detecting a change, user is **alerted** with the key path, change **type** (add/modify/delete), and an option to **revert**

Central Keys	
Run/RunOnce	Runs programs on login
TimeZoneInfo	Controls date & time settings (for testing)
Internet Explorer/Main	Used by LoveBug to download .exe

Table created by finalists using Canva



Flowchart created by finalists using LucidChart



Graphic created by finalists using Canva

RESULTS

Design Goals:

- RegiMon met **3 of 3** primary design goals:
- Real-time detection: RegNotifyChangeKeyValue() successfully detected all changes with **no polling delay**.
- **Undo** functionality: successful across testing
- Lightweight system tray app: The production has an **installer** .exe and runs as a tray application with **zero CPU** overhead while idle.
- One partially unmet secondary goal: the passive notification system (background alerts without manual UI interaction) remains incomplete.
- RegNotifyChangeKeyValue() compensated for time lost to early setbacks.

Limitations

- Notification system and tray icon not yet fully stabilized: alerts require some manual interaction with the UI.
- Testing scope limited by environment constraints; only Love Bug tested as a malware sample.

CONCLUSION

- Addressed research gap with continuous monitoring
- Can be used in security labs, classroom research, sandboxed VMs, and endpoint monitoring.
- Aligns with Dolan-Gavit et al.'s (2008) finding that volatile memory analysis is more efficient and revealing than on-disk registry.
- App remained self-contained outside of installation
- Better suited for proactive threat detection rather than post-incident forensics (Dolan-Gavit et al., 2008; Morgan, 2008)

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